# HSF302 - Retrieval Augmented Generation (RAG) Architecture - Page 1

Chapter 1: RAG Architecture and Vector Search Overview
Retrieval-Augmented Generation (RAG) grounds Large Language Models on verified course textbook documents to eliminate hallucinations.
1. Document Chunking: Lecture slides and PDF textbooks are extracted using Apache PDFBox and partitioned into semantic chunks of 300 to 500 words with sentence boundaries preserved.
2. Semantic Retrieval: Cosine similarity and keyword overlap score relevant chunks against the student query.
3. Context Injection: Top matching chunks are injected into the Gemini System Prompt with strict citation constraints including source file name and page numbers.